

■ Machine Learning Formula Handbook (Running PDF)

■■ Date: 08/12/2025

Linear Regression:

$$y = mx + b$$

$$\text{Multiple Linear Regression: } y = m_1 \cdot x_1 + m_2 \cdot x_2 + \dots + b$$

Cost Function (MSE):

$$J(m,b) = (1 / 2m) * \sum (y_{\text{pred}} - y)^2$$

Gradient Descent Update Rules:

$$m = m - \alpha * \partial J / \partial m$$

$$b = b - \alpha * \partial J / \partial b$$

$$\text{Derivative example: } d(x^2)/dx = 2x$$

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Entropy:

$$\text{Entropy} = - \sum (p * \log_2(p))$$

Gini Impurity:

$$\text{Gini} = 1 - \sum (p^2)$$

Sigmoid Function:

$$\sigma(z) = 1 / (1 + e^{(-z)})$$

Min-Max Scaling:

$$X' = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$